THOMSON COURSE TECHNOLOGY

Guide to Network Defense and Countermeasures Third Edition

Chapter 2

TCP/IP

Objectives

- Explain the fundamentals of TCP/IP networking
- Describe IPv4 packet structure and explain packet fragmentation
- Describe Internet Protocol version 6 (IPv6)

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The OSI Model and TCP/IP Protocols

- Transmission Control Protocol/Internet Protocol (TCP/IP) is a suite of many protocols for transmitting information from point to point on a network
  - Often referred to as a “stack”
- This section covers:
  - Open System Interconnection (OSI) model
  - IP addressing
  - subnetting

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The OSI Model, Part 1

- OSI reference model: divides the communication functions used by two hosts into seven separate layers
- TCP/IP has its own stack of protocols that correspond to these layers

OSI	TCP/IP stack
Application	HTTP DNS DHCP
Presentation	FTP SNMP Telnet
Session	IMAP SMTP POP
Transport	TCP UDP
Network	IP ICMP IGMP ARP RIP OSPF
Data Link	Device Drivers
Physical	Network Adapter

Table 2-1 The OSI model and the subprotocols of the TCP/IP stack

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The OSI Model, Part 2

- TCP/IP subprotocols are services that support a number of network functions:
  - HTTP (Hypertext Transfer Protocol)
  - DNS (Domain Name System)
  - DHCP (Dynamic Host Configuration Protocol)
  - FTP (File Transport Protocol)
  - SNMP (Simple Network Management Protocol)
  - Telnet
  - IMAP, SMTP, POP

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TCP/IP Addressing, Part 1

- IP addresses are a method used to identify computers
  - Processed at the Network layer of the OSI model
  - Most common in use conform to Internet Protocol version 4 (IPv4)
    - 32-bit address divided into four groups called octets
    - Each octet contains 8 bits of data
  - In binary, an IP address looks like:
    - 10000000.00100110.00101100.11100010
  - Binary is converted to dotted decimal notation
    - 192.168.10.1

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TCP/IP Addressing, Part 2

- IP address components
  - Network identifier – shared among computers in a network segment
  - Host address – unique to each computer on the network segment
- Subnet mask – used to identify which part of the IP address is the network identifier and which is the host identifier
- Attackers can gain access to a network by determining IP addresses of computers
- IP addresses need to be concealed to prevent certain attacks

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TCP/IP Addressing, Part 3

- If an attacker can find a PC's IP address, they can run a port scan to look for open ports to exploit
- To hide addresses, use **Network Address Translation (NAT)**
  - Translates private network's internal addresses into external addresses that can be used on the public Internet
    - Private network's internal addresses are not routable on the Internet
- Today IP addresses are in short supply, so Internet Protocol version 6 (IPv6) is being implemented

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Address Classes

- IPv4 addresses are separated by classes
  - Class is determined by the number of its networks compared to number of hosts
  - Example: a Class A address uses 8 bits for the network portion and 24 bits for the host portion

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IP Address Classes

Class	First octet decimal range	Default subnet mask	Purpose
Class A	1-126 127-x.x.x is reserved; the address 127.0.0.1 is used to indicate the local system's TCP/IP implementation	255.0.0.0	Large corporations and governments
Class B	128-191	255.255.0.0	Medium networks
Class C	192-223	255.255.255.0	Small networks
Class D	224-239	N/A	Multicasting
Class E	240-254	N/A	Experimentation

Table 2-2 IP address classes

Private IP Address Ranges

- Private addresses are needed so that organizations can build internal infrastructures
  - Public IP addresses require registration and a fee for each address
  - Private addressing scheme eliminates the need to purchase addresses for every group of machines

Network address	Subnet mask	First valid host address	Last valid host address	Broadcast address
10.0.0.0	255.0.0.0	10.0.0.1	10.255.255.254	10.255.255.255
172.16.0.0	255.240.0.0	172.16.0.1	172.31.255.254	172.31.255.255
192.168.0.0	255.255.0.0	192.168.1.1	192.168.255.254	192.168.255.255

Table 2-3 Private IP address ranges

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Subnetting, Part 1

- Address classes already have network id octets by default
  - Class A uses first octet
  - Class B uses first two octets
  - Class C uses first three octets
- Default Class B has 16 bits available for hosts
  - This means a Class B network can have more than 65,000 host addresses
  - Some of host bits can be used to identify the network
  - Creates smaller subnetworks with fewer hosts

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Subnetting, Part 2

- Subnetting can be used for:
  - Mirroring the organization's physical layout
  - Mirroring the organization's administrative structure
  - Planning for future growth
  - Reducing and controlling network traffic
  - Increasing network security
- If all users with similar security and access needs are grouped into a single subnet, the entire group can be managed instead of managing each user separately

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Subnetting, Part 3

- Subnetting
  - Borrow bits from host portion of IP address
  - Number of borrowed bits determines how many subnets and hosts are available
  - At least two bits must be available for hosts
    - Up to 14 bits can be borrowed for a Class B network

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Subnetting, Part 4

Subnet	Number of subnetworks	Usable hosts per subnet
255.255.128.0	2	32766
255.255.192.0	4	16384
255.255.224.0	8	8190
255.255.240.0	16	4094
255.255.248.0	32	2046
255.255.252.0	64	1022
255.255.254.0	128	510
255.255.255.0	256	254
255.255.255.128	512	126
255.255.255.192	1024	62
255.255.255.224	2048	30
255.255.255.240	4096	14
255.255.255.248	8192	6
255.255.255.252	16384	2

Table 2-4 Class B subnetting

Subnetting, Part 5

Binary digit	1	1	1	1	1	1	1	1
Decimal equivalent	128	64	32	16	8	4	2	1

Table 2-5 Binary-to-decimal values

Subnetting, Part 6

- Subnetting a Class C address example:
  - Network address: 199.1.10.0
  - Default subnet mask: 255.255.255.0
  - Selected mask: 255.255.255.224
  - Mask in binary: 11111111.11111111.11111111.11100000
    - Last masked digit occupies the binary value of 32
    - Starting with network address, increment by 32 until you reach the mask's number (224)

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Subnetting, Part 7

Subnet address	Valid host address range	Broadcast address for subnet
199.1.10.0	199.1.10.1-199.1.10.30	199.1.10.31
199.1.10.32	199.1.10.33-199.1.10.62	199.1.10.63
199.1.10.64	199.1.10.65-199.1.10.94	199.1.10.95
199.1.10.96	199.1.10.97-199.1.10.126	199.1.10.127
199.1.10.128	199.1.10.129-199.1.10.158	199.1.10.159
199.1.10.160	199.1.10.161-199.1.10.190	199.1.10.191
199.1.10.192	199.1.10.193-199.1.10.222	199.1.10.223
199.1.10.224	199.1.10.225-199.1.10.254	199.1.10.255

Table 2-6 Subnetting example

Variable Length Subnet Masking

- Networks that do not have a large number of available IP addresses use variable length subnet masking (VLSM)
  - Involves applying masks of varying sizes to the same network
  - Creates subnets within subnets
  - Often used to secure stub networks (only have one connection to any other network

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Classless Interdomain Routing

- Classless Interdomain Routing (CIDR) – specifies the number of masked bits in an IP address/subnet mask combination
- Example:
  - A network address of 192.168.6.0 with a subnet mask of 255.255.255.224 would have a CIDR notation of 192.168.6.0/27
- CIDR overcomes limitations of default subnet masks so that unused addresses do not go to waste

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Unicasting, Multicasting, and Broadcasting

- Unicast transmission: one packet is sent from one server to each client computer individually
- Multicast transmission: server can treat several computers as a group and send one transmission that reaches all of them
  - Example: streaming video presentation
- Broadcast transmission: sent to all nodes on a specific network
  - Flooded broadcasts: sent to any subnet
  - Directed broadcasts: sent to a specific subnet

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Examining Internet Protocol Version 4 (IPv4)

- IP datagrams
  - Portion of the packet that is responsible for routing through networks
  - Processed at the Network layer of the OSI model
  - Complete message is transmitted using multiple datagrams
  - Contain information about source and destination IP addresses, control settings, and data
  - Divided into different sections
    - Primary subdivisions are header and data
    - Some packets have a footer (or trailer) that indicates the end of a packet or error checking (CRC)

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IP Header Structure, Part 1

- Part of an IP packet that computers use to communicate
- IP header plays an important role in terms of network security and intrusion detection
- Contains a number of fields and is similar to a TCP header

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IP Header Structure, Part 2

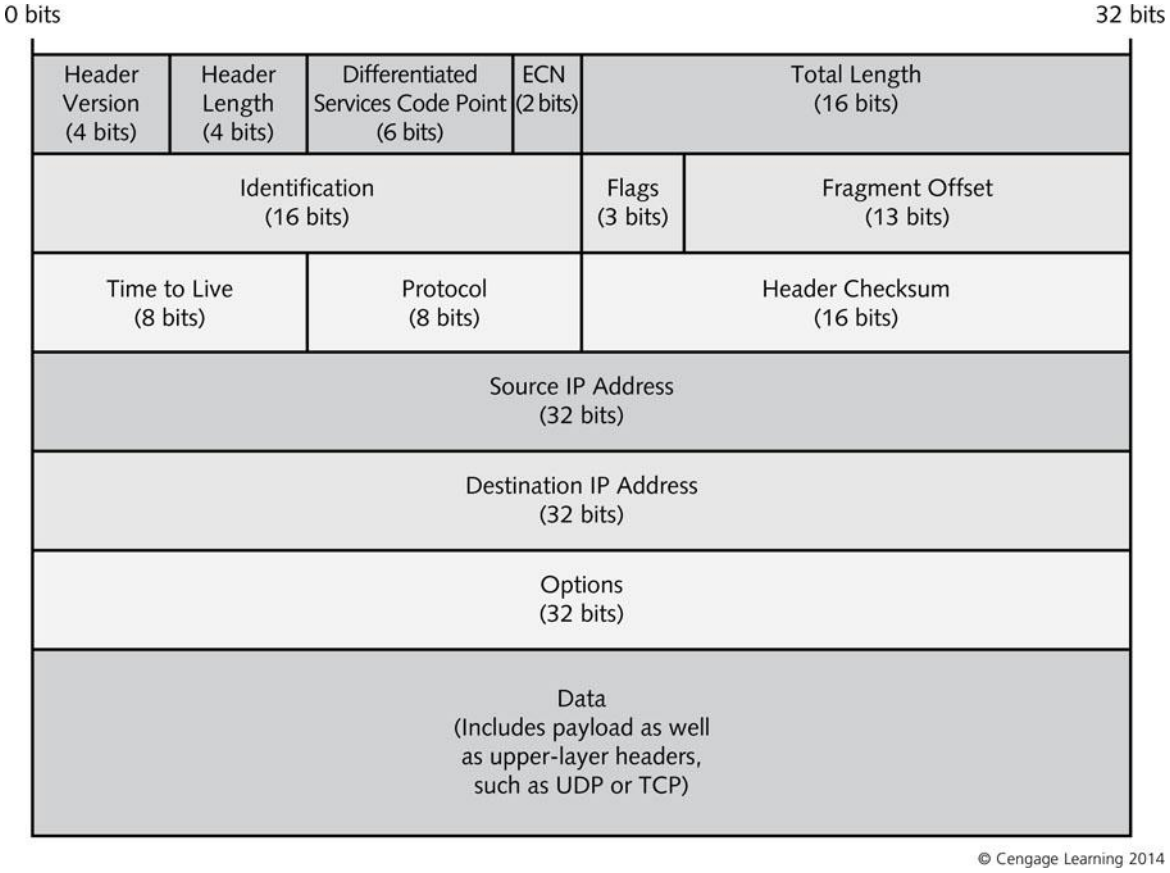

Figure 2-1 IP header structure

IP Header Structure, Part 3

- Most network operating systems (NOSs) have a type of built-in or add-on program to monitor network activity
- Most administrators prefer third-party applications for their versatility and extra features
  - Wireshark (formerly Ethereal) is an example

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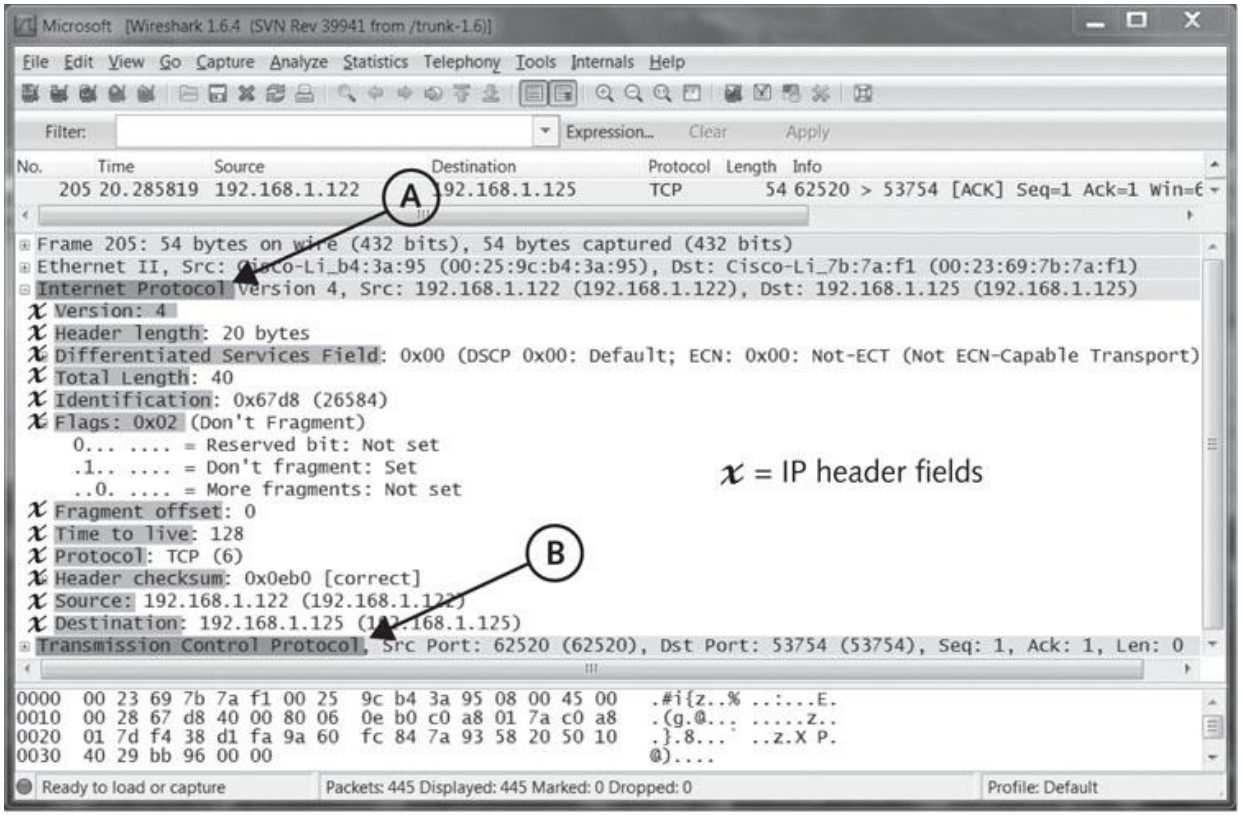
IP Header Structure, Part 4



Source: Wireshark

Figure 2-2 IP header structure as seen in a Wireshark packet capture

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ICMP Messages

- Internet Control Message Protocol (ICMP) used to assist with troubleshooting communication problems
  - Ping command uses ICMP to check whether a remote host has connectivity
- Processed at the network layer of the OSI model
- Firewalls or packet filters can be configured to accept or deny certain ICMP packets through the network
  - Some ICMP packets could be used as part of an attack

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ICMP Types

ICMP type	Name	ICMP type	Name
0	Echo Reply	17	Address Mask Request
3	Destination Unreachable	18	Address Mask Reply
4	Source Quench	30	Traceroute
5	Redirect	31	Datagram Conversion Error
6	Alternate Host Address	32	Mobile Host Redirect
8	Echo	33	IPv6 Where-Are-You
9	Router Advertisement	34	IPv6 I-Am-Here
10	Router Selection	35	Mobile Registration Request
11	Time Exceeded	36	Mobile Registration Reply
12	Parameter Problem	37	Domain Name Request
13	Timestamp	38	Domain Name Reply
14	Timestamp Reply	39	SKIP
15	Information Request	40	Photuris
16	Information Reply	1-2, 7, 19-29, 41-252	Unassigned or Reserved

Table 2-7 ICMP types

TCP Headers

- TCP/IP packets may also contain TCP headers
  - TCP headers are processed at the Transport layer of OSI model
  - TCP portion of a packet is called TCP segment
  - Flags section of a TCP header are important:
    - You can specify them when you create packet-filtering rules

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TCP Header Structure

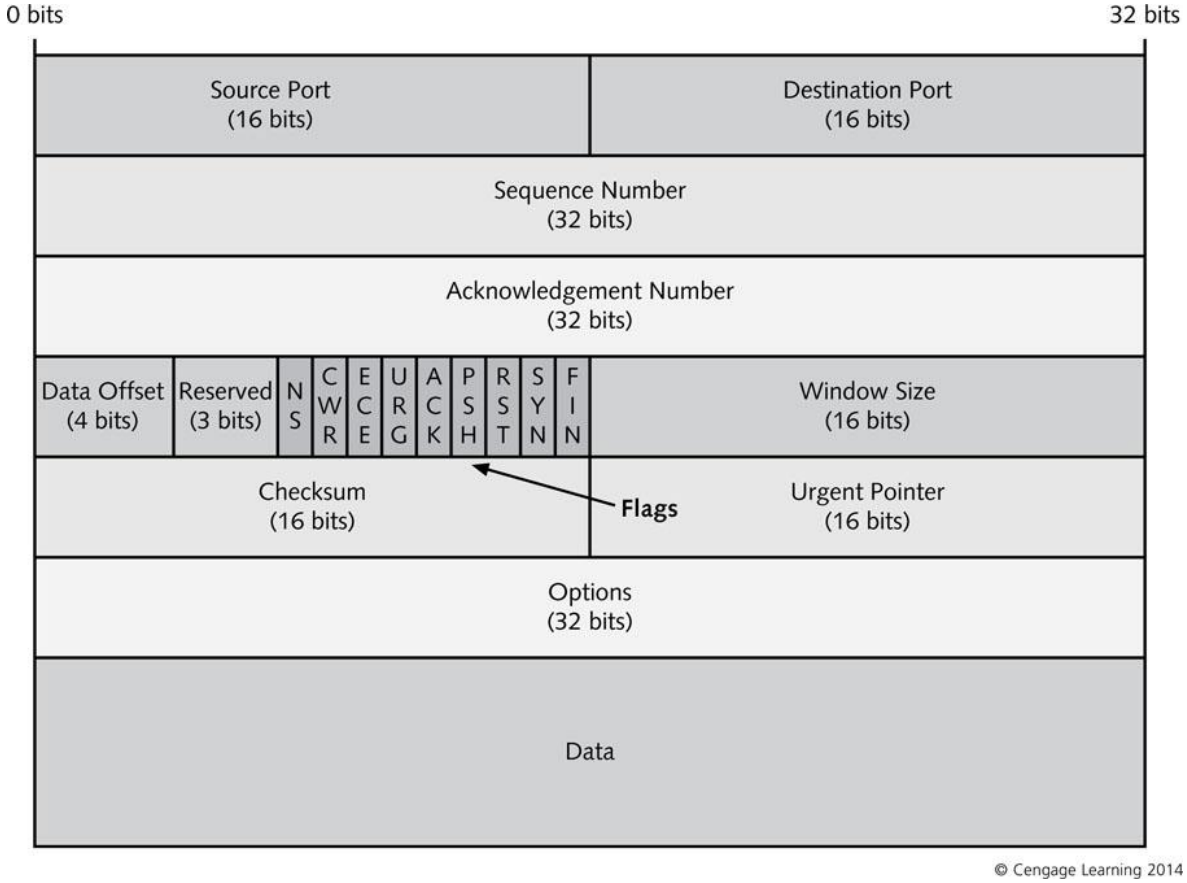

Figure 2-3 TCP header structure

UDP Headers

- User Datagram Protocol (UDP): provides a transport service for IP
  - Processed at Transport layer of OSI model
  - Considered unreliable because it is connectionless
    - UDP packet does not contain sequence or acknowledgement numbers that enable TCP to guarantee delivery
  - Much faster than TCP
  - Used for broadcasting messages or for protocols that do not require the same level of service as TCP
  - Attackers can scan for open UDP services

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UDP Header Structure

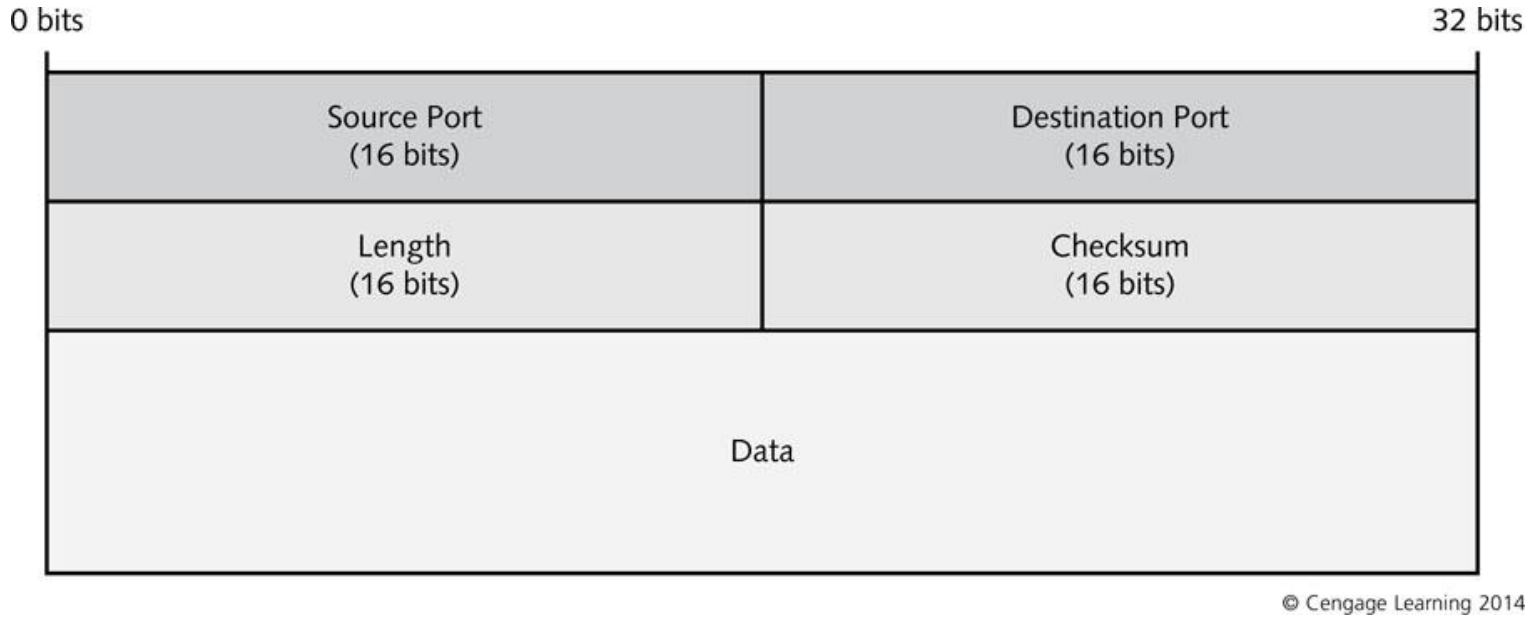

Figure 2-5 UDP header structure

Packet Fragmentation

- Originally developed to allow large packets to pass through routers with frame size limitations
  - Routers divide packets into multiple fragments and send them along the network
- Fragmentation creates security problems
  - Port numbers appear only in fragment 0
  - Fragments 1 and higher pass through filters without being scrutinized
    - Attacker can modify the IP header to make all fragment numbers start at 1 or higher
  - Configure firewall to drop all fragmented packets

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The TCP Life Cycle and the TCP Three-Way Handshake, Part 1

- Establishing connection-oriented communication using a three-way handshake:
  - Host A sends an initial sequence number in its first packet to Host B
    - Called a SYN packet
  - Host B receives SYN packet - responds with SYN ACK with an initial sequence number for Host B
    - Includes an acknowledgement number that is one more than the initial sequence number
  - Host A sends an ACK packet to Host B
    - Increases Host B's sequence number by one

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TCP Three-Way Handshake: SYN ACK

Sending computer	Host A
Source TCP port	26077
Destination TCP port	80
Sequence Number	50088
Acknowledgement number	0
Flags	SYN

Table 2-8 TCP three-way handshake: SYN

Sending computer	Host B
Source TCP port	80
Destination TCP port	26077
Sequence number	79995
Acknowledgement number	50089
Flags	SYN ACK

Table 2-9 TCP three-way handshake: SYN ACK

TCP Three-Way Handshake: ACK

Sending computer	Host A
Source TCP port	26077
Destination TCP port	80
Sequence number	50089
Acknowledgement number	79996
Flags	ACK

Table 2-10 TCP three-way handshake: ACK

The TCP Life Cycle and the TCP Three-Way Handshake, Part 2

- Sliding window size: determines the number of packets that can be sent before ACKs must be received
  - Controls the flow and efficiency of communications
  - Sender controls size of sliding window
- FIN flag is set when either side is ready to end the session
  - Station that receives the initial flag sends a response packet with the ACK flag and its own FIN flag set to acknowledge receipt and to show it is ready to end the session

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Summary of the TCP Three-Way Handshake

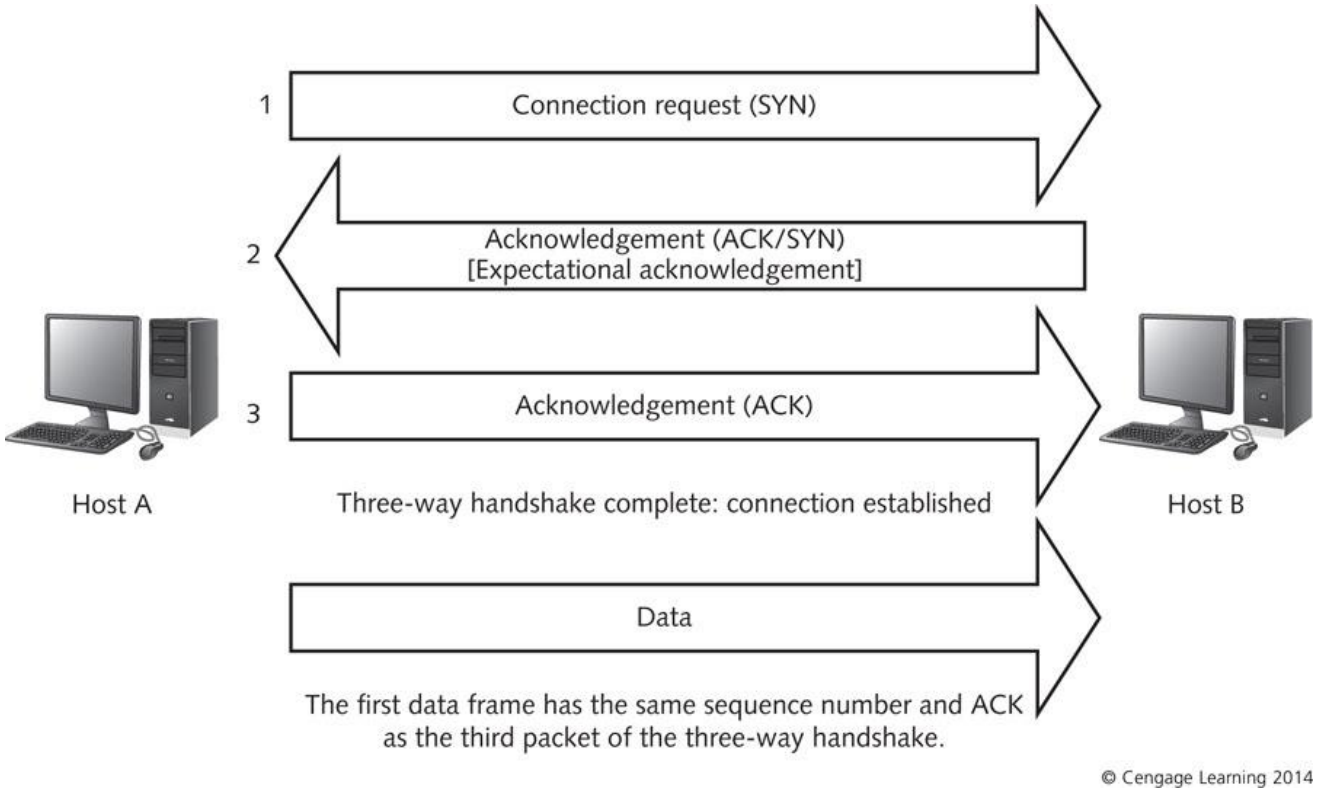

Figure 2-7 Summary of the TCP three-way handshake

Domain Name Service

- DNS servers translate fully qualified domain names (FQDNs) to IP addresses
- DNS can be used to block unwanted communications
  - Administrators can block Web sites containing offensive content
- DNS attacks
  - Buffer overflow
  - Zone transfer
  - Cache poisoning

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Internet Protocol Version 6 (IPv6)

- IPv6 addresses the many limitations of IPv4
  - IPv6 has a larger address space of 128 bits
  - Routing tables need only the entries of other routers that are directly connected to them
  - IPv6 has integrated support for security called IPsec
  - Network Address Translation (NAT) is not needed
    - NAT has security problems
  - IPv6 can determine its own settings based on two different models:
    - Stateful autoconfiguration
    - Stateless autoconfiguration

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IPv6 Core Protocols, Part 1

- IPv6 has major differences to IPv4 in its core architecture and functions
  - It is a connectionless, unreliable datagram protocol used mainly for addressing and routing packets
- IPv6 datagram consists of the IPv6 header and IPv6 payload
  - Header is made up of IPv6 base header and optional extension headers

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IPv6 Header Structure

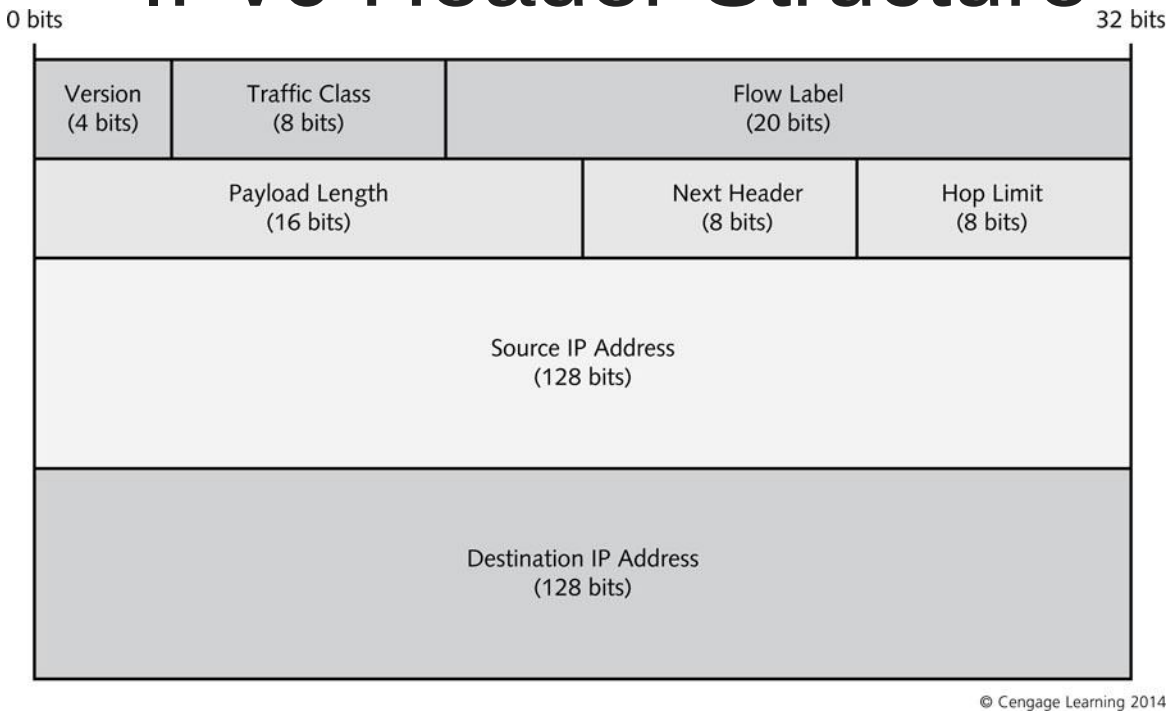

Figure 2-8 IPv6 header structure

IPv6 Core Protocols, Part 2

- Extension headers are not normally found in a typical IPv6 packet
  - If needed, the sending host adds appropriate header
  - IPv6 extension headers:
    - Hop-by-Hop Options
    - Destination Options
    - Routing
    - Fragment
    - Authentication
    - Encapsulating Security Payload (ESP)

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Internet Control Message Protocol for IPv6

- ICMPv6 messages are grouped into two classes:
  - Error messages: 0-127
  - Informational messages: 128-255
- ICMPv6 messages is preceded by an IPv6 header
  - Sometimes by extension headers
- Type field contains the value for a type of message

Message type	Type
Destination Unreachable	1
Packet Too Big	2
Time Exceeded	3
Parameter Problems	4
Echo Request	128
Echo Reply	129

Table 2-11 Common ICMPv6 message type codes

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ICMPv6 Features

Message type	Code
Multicast Listener Discovery (MLD)	MLD replaces IGMPv3 (Internet Group Message Protocol), which is used for controlling multicasts in IPv4. MLD is a series of three ICMPv6 messages used to manage subnet multicast membership. (See Table 2-13.)
Neighbor Discovery (ND)	ND replaces ARP, ICMPv4 Router Discovery, and ICMPv4 Redirect and has other functions, including prefix detection, duplicate address detection, and automatic address configuration. ND is a series of five ICMPv6 messages used to manage node-to-node communications on a link. (See Table 2-14.)

Table 2-12 ICMPv6 features

Multicast Listener Discovery, Part 1

- Multicasts: connectionless delivery of information to multiple subscribers at the same time
  - Has a single stream on any link instead of one stream per recipient
- IP multicast traffic is sent to a single address but is processed by all members of a multicast group
  - Hosts listening on a specific multicast address are part of the multicast group
  - Group membership is dynamic
  - Members can be on different subnets

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Multicast Listener Discovery, Part 2

- **Multicast Listener Discovery (MLD)** enables IPv6 routers to discover multicast listeners and decide which multicast addresses are of interest to nodes

MLD message type	Description
Multicast Listener Query	Multicast routers send queries to poll a network segment for group members. Queries can be general, can request membership for all groups, or can be used for a specific group.
Multicast Listener Report	This message is sent by a host when it joins a multicast group or in response to a Multicast Listener Query.
Multicast Listener Done	This message is sent by a host when it leaves a host group and is the last member of that group on the network segment.

Table 2-13 Multicast Listener Discovery message types

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Neighbor Discovery

- Neighbor Discovery (ND): new IPv6 protocol that replaces ARP, ICMP Router Discovery and ICMP Redirect in IPv4
  - Uses ICMPv6 messages to manage node-to-node communications
- Five different types of ICMP messages:
  - Router Solicitation
  - Router Advertisement
  - Neighbor Solicitation
  - Neighbor Advertisements
  - Redirect

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IPv6 Neighbor Discovery Functions

Process	Description
Router discovery	Discovers neighboring routers
Prefix discovery	Discovers local network prefixes (equivalent to IPv4 network addresses)
Parameter discovery	Discovers additional parameters, such as MTU (Maximum Transmission Unit) size for the network segment and default hop limit for outbound packets
Address autoconfiguration	Automatically configures addresses
Address resolution	Resolves a neighboring node's address to its MAC address
Next-hop determination	Determines the next-hop node address, which is typically the final destination or a router on the network segment
Neighbor unreachability detection	Determines whether neighboring hosts or routers are no longer available
Duplicate address detection	Determines that an address considered for use is not already in use by a neighboring node
Redirect function	Determines the process by which a router informs a host of a better first-hop IPv6 address to reach a destination

Table 2-14 IPv6 Neighbor Discovery functions

IPv6 Addressing, Part 1

- IPv6 addresses:
  - 128 bits long and use the hexadecimal numbering format
  - Consist of eight hex groups separated by colons
    - Each hex group contains a 16-bit value
- Examples:
  - 4EDC:0000:7654:3210:F3DC:BA98:7654:AB1F
  - Including leading zeros is not necessary
    - 1080:0:0:0:8:800:200C:417A
  - Can replace consecutive zeros with a double colon
    - 1080::8:800:200C:417A

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IPv6 Addressing, Part 2

- Unicast addressing: used for one-to-one communication (between two host or two routers)
- Scopes of unicast addresses:
  - Global unicast address: public addresses routable on the Internet
  - Site-local unicast address: similar to private IPv4 addresses
  - Unique local IPv6 unicast address: replacing site-local unicast address
  - Link-local unicast address: used by hosts to communicate with other hosts on same network

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IPv6 Addressing, Part 3

- Multicast addresses: used for one-to-many communications
  - Always begin with FF in the first byte
- Anycast addresses: used for one-to-one or one-to-many communications
  - Created automatically when a unicast address is assigned to more than one interface
  - Offers flexibility in providing services
  - Currently only used by routers but will expand as technology becomes widespread

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IPv6 Configuration

- Microsoft OSs since Windows XP SP1 have built-in support for IPv6 support
  - Support stateless autoconfiguration
- A link-local address is assigned to every Ethernet interface during startup
  - Assigned automatically based on receipt of IPv6 Router Advertisement messages
  - Must have a correctly configured IPv6 capable router on network segment

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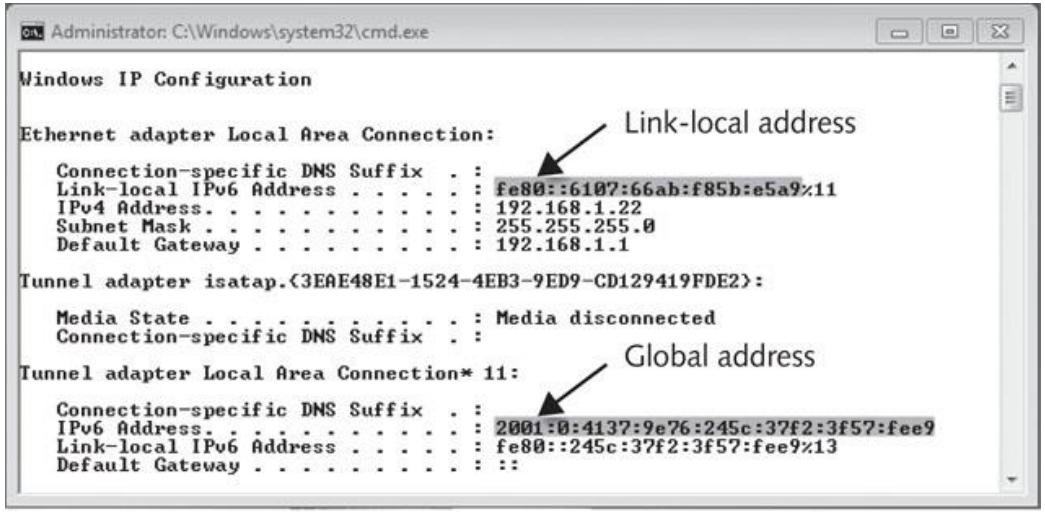
IPv6 Utilities, Part 1

- **Ipconfig**: shows IPv6 configuration details
  - Can also use the command with IPv4



Source: Used with permission of Microsoft Corporation

Figure 2-10 Using the ipconfig command

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IPv6 Utilities, Part 2

- **Netstat**: display system's routing tables by using netstat –*r* command
  - Netstat -*n* option shows current sessions with the associated port numbers
  - Netstat –*ps* IPv6 option displays detailed statistics on IPv6 activity since the last boot
- **Netsh**: command-line scripting tool on Windows systems that allows troubleshooting and configuration of network interfaces

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Summary, Part 1

- TCP/IP is a suite of protocols for transmitting information from point to point on a network
- TCP and UDP map to the Transport layer and IPv4, IPv6, ICMP, and ICMPv6 map to the Network layer of the OSI model
- IP addresses most commonly used on the Internet conform to IPv4
- You must understand the normal configuration of fields in IP, TCP, and UDP headers to recognize and filter unwanted or malicious traffic

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Summary, Part 2

- Fragmentation of IP packets allows large packets to pass through routers with frame size limits
- DNS translates fully qualified domain names into IP addresses
- TCP three-way handshake establishes a reliable connection between two points
- IPv6 was designed to address problems with IPv4
- IPv6 is a connectionless, unreliable protocol used mainly for addressing and routing packets
- ICMP is used for reporting errors and diagnostics

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Summary, Part 3

- MLD enables IPv6 routers to discover multicasts
- IPv6 uses ND to perform tasks that ARP, ICMP Router Discovery and ICMP Redirect handled
- Hexadecimal numbering format makes IPv6 addresses manageable
- IPv6 uses three types of addresses: unicast, multicast, and anycast
- You can monitor and configure IPv6 using tools such as Ipconfig, Netstat, and Netsh

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